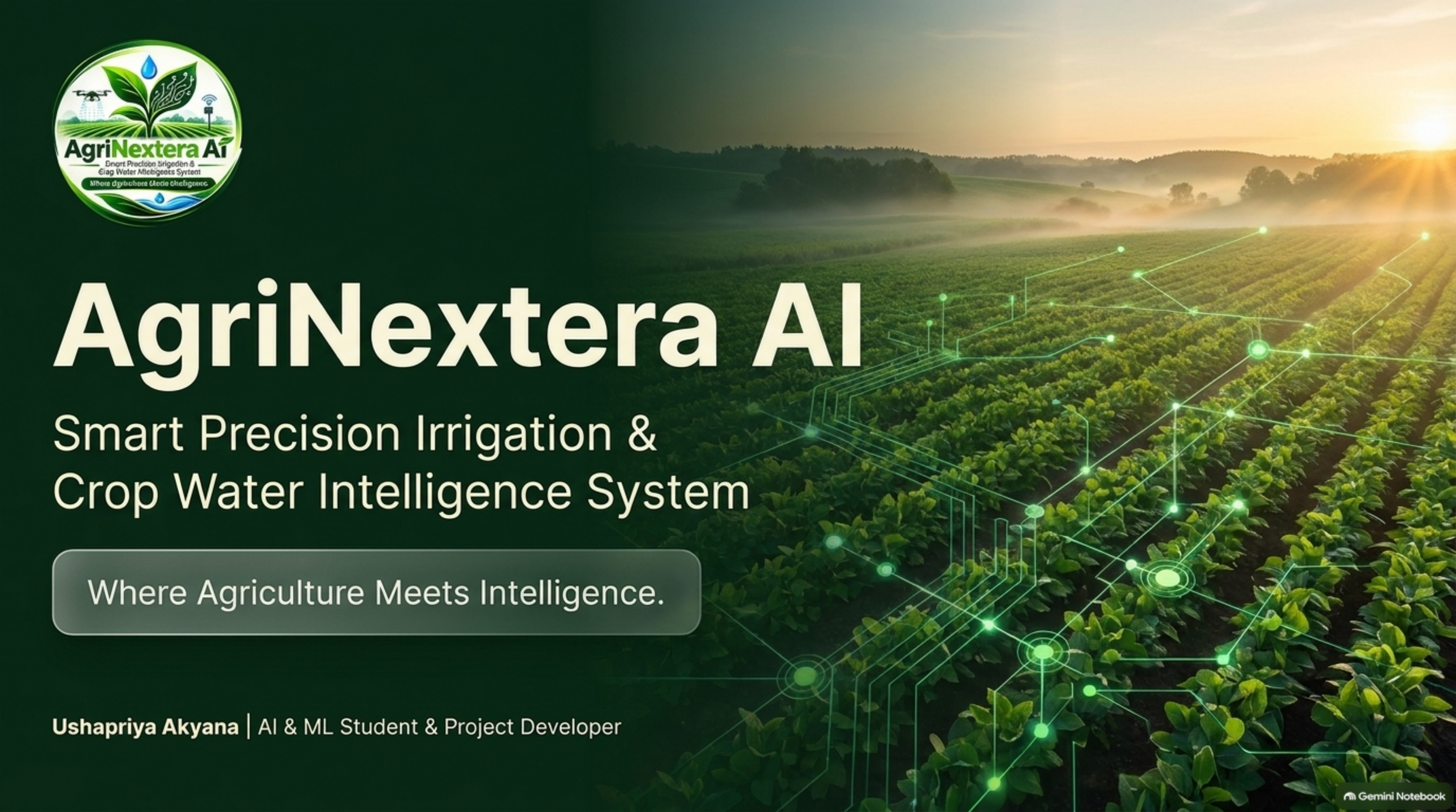


AgriNextera AI

Smart Precision Irrigation & Crop Water Intelligence System

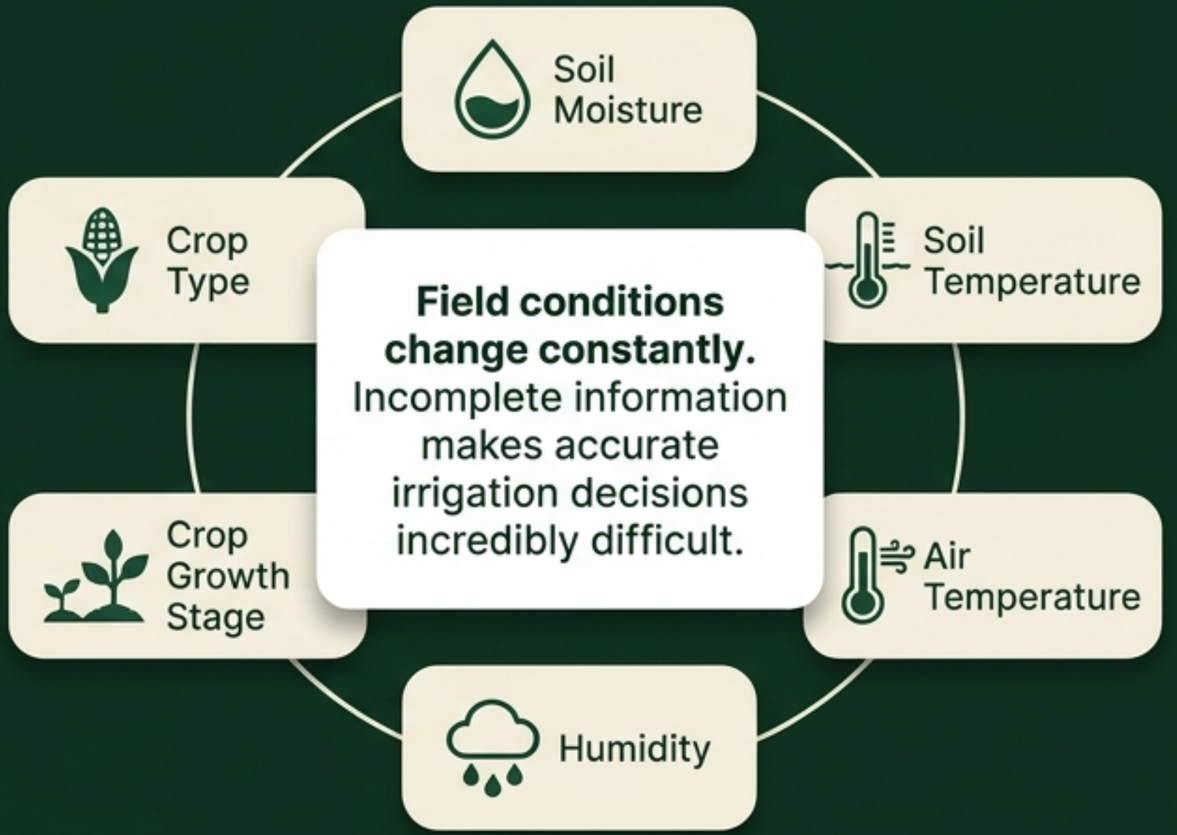
Where Agriculture Meets Intelligence.

Ushapriya Akyana | AI & ML Student & Project Developer

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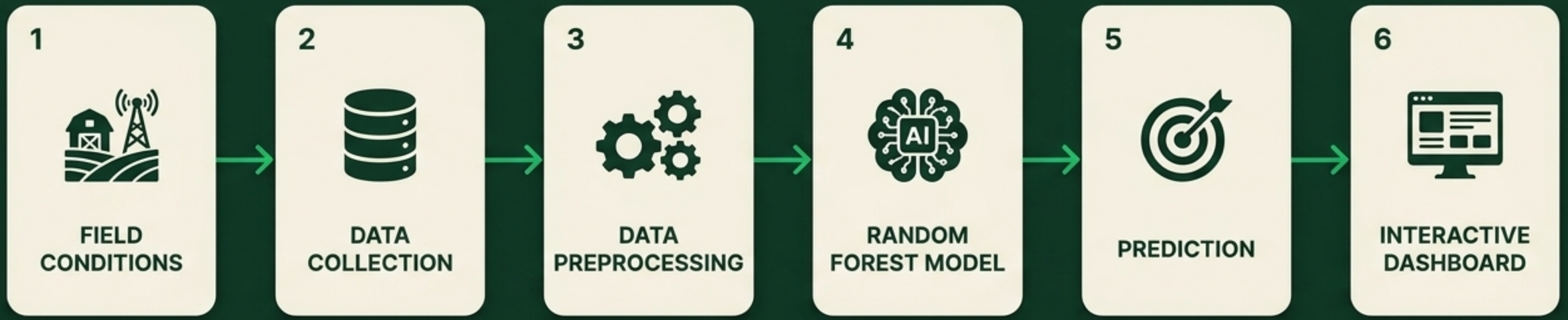
Every Drop Matters. Every Decision Counts.



What if field data could become smarter irrigation insights?

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Introducing AgriNextera AI



An AI-powered prototype that analyzes complex field variables and predicts irrigation-related classes, delivering real-time probabilities directly to the user.

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From Field Data to Machine Intelligence

The Dataset

150 records total.

11 original columns reduced to **7** meaningful features.

0 missing values | **0** duplicate rows.

Inputs & Target

Features:

CropType, CropDays, Soil Moisture, Soil Temp, Temp, Humidity.

Target:

Irrigation (Y/N), represented as Class 0 and Class 1.

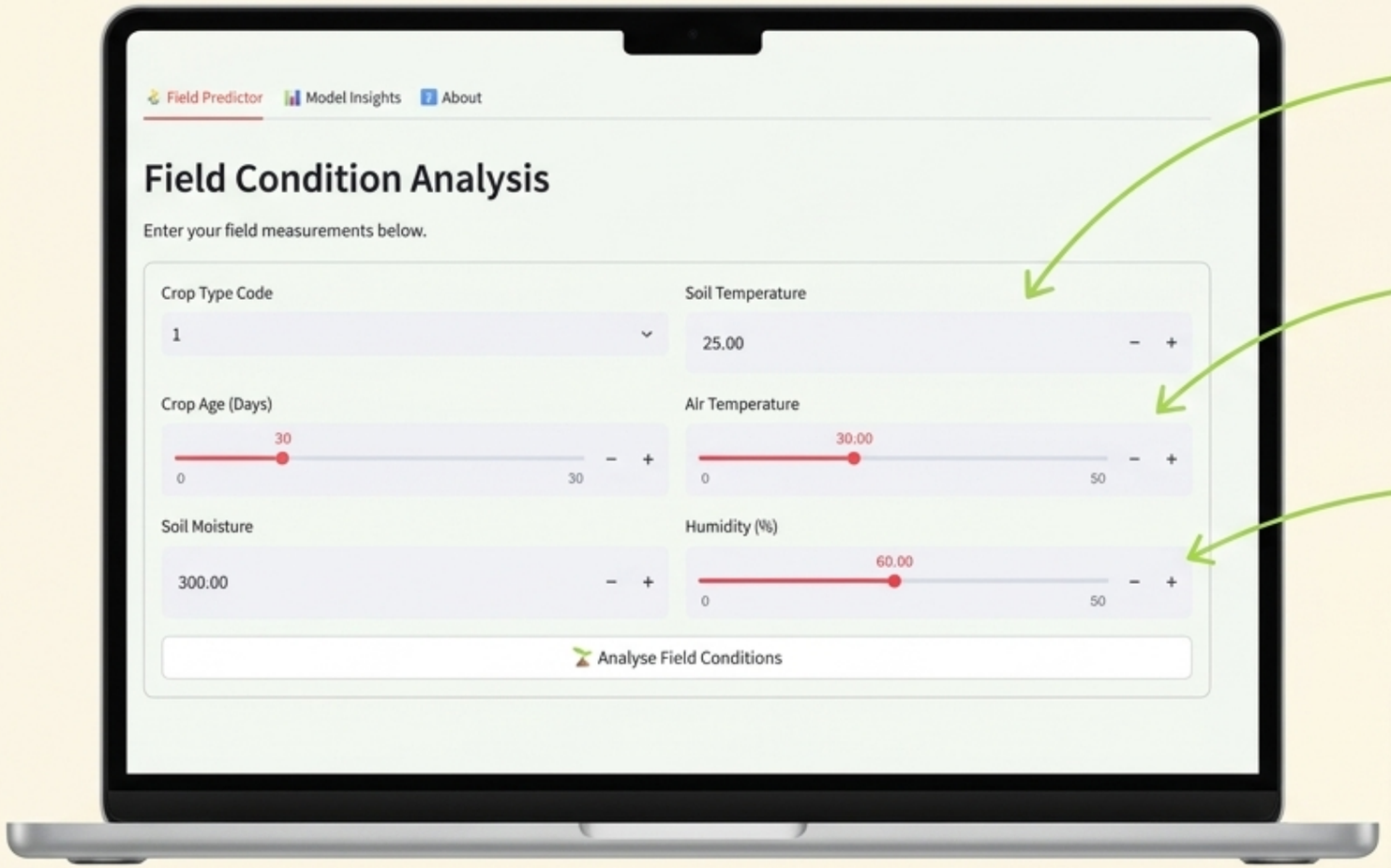
Note: Numerical target labels (0/1) are from the dataset and have not been independently verified for real-world agronomic meaning.

The ML Engine

- **Algorithm:** Random Forest Classifier (100 trees, Random state: 42).
- **Weighting:** Balanced class weighting.
- **Split:** Stratified 80:20 train-test split.
- **Validation:** Holdout testing & 5-fold stratified cross-validation.

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Not Just an Idea. A Working AI Application.



Real-time input sliders for all 6 environmental variables.

Instant model predictions and probability distributions.

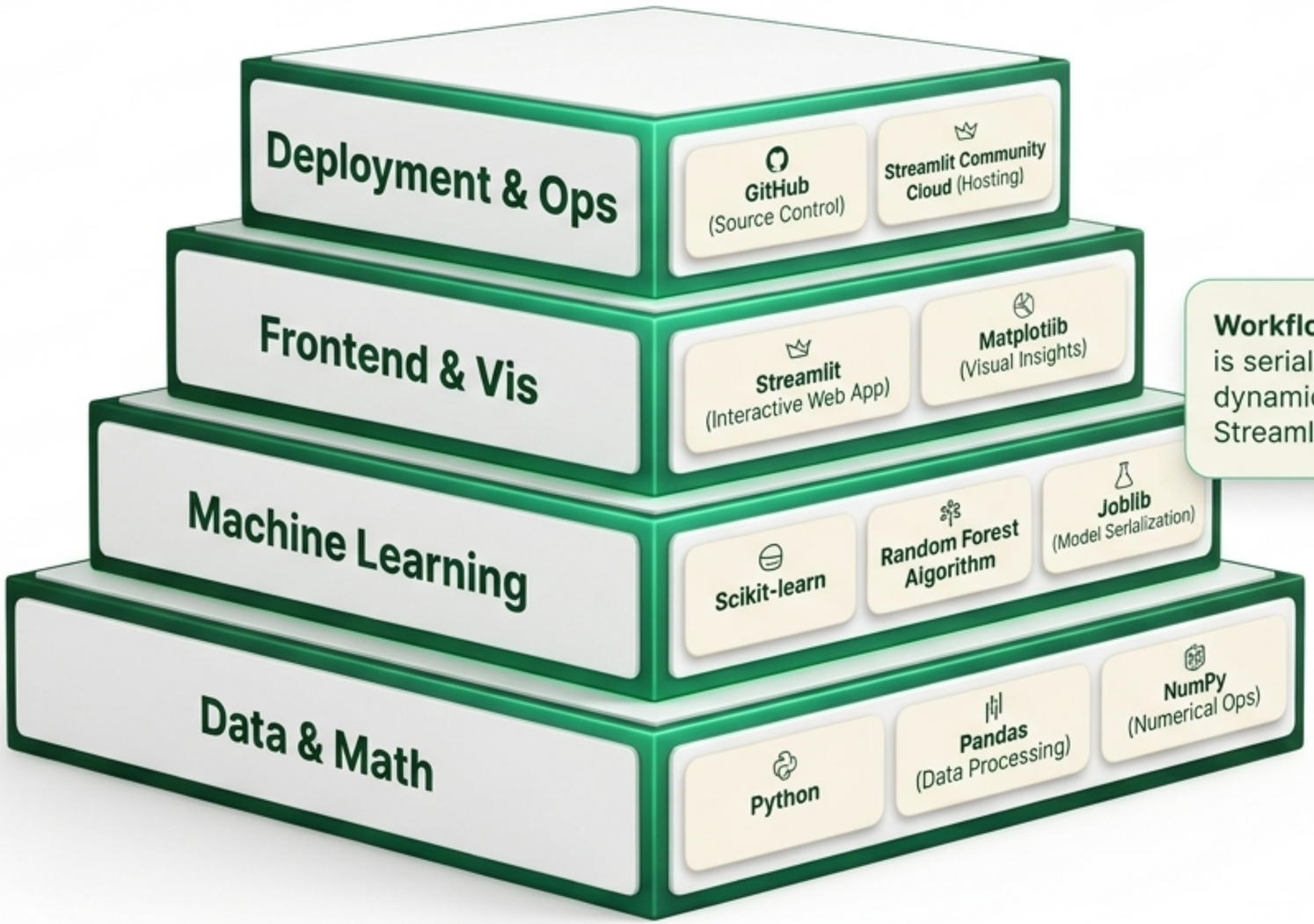
Integrated 'Model Insights' and 'About' tabs.

EXPLORE LIVE DEMO

https://agrinextera-ai.streamlit.app/

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The Intelligence Stack



Workflow Note: The trained model is serialized via Joblib and loaded dynamically by the deployed Streamlit application.

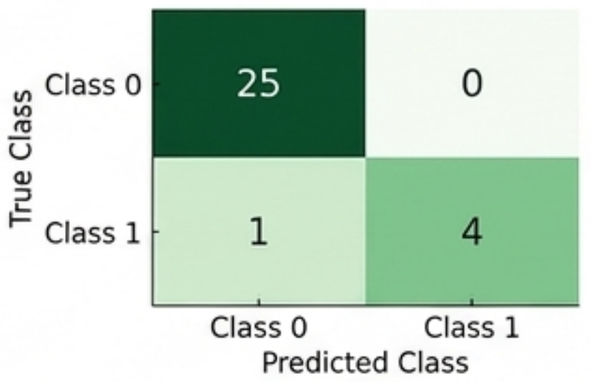
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Measuring Model Performance

96.67%

Holdout Accuracy (29 of 30 correct predictions).

Confusion Matrix



Cross-Validation

Accuracy: **94.0%**	Precision: **0.874**
Recall: **0.800**	F1-Score: **0.821**

Holdout test accuracy: 96.67%

Test records: 30

Correct predictions: 29 out of 30

Cross-validation average accuracy: 94%

Class-wise Performance

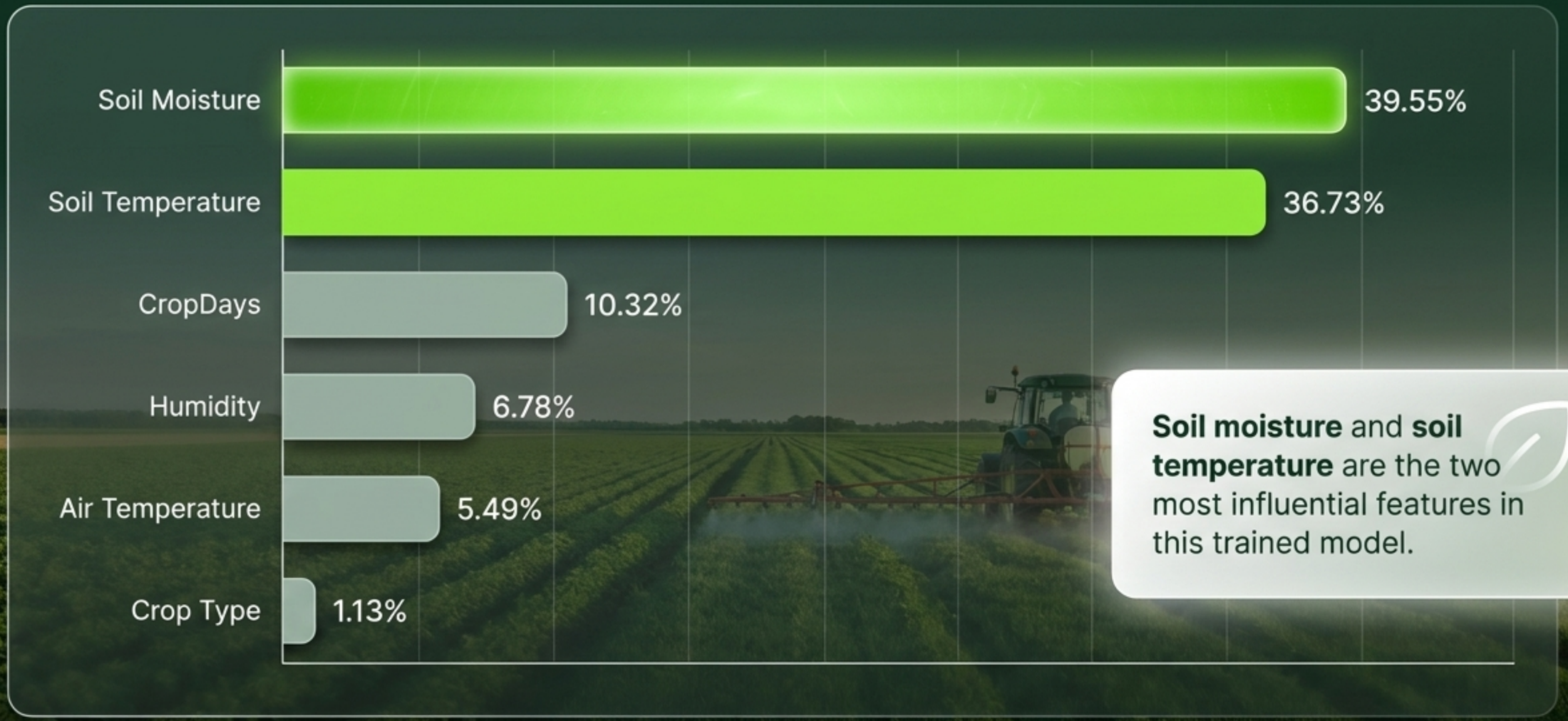
Class 0 (F1: 0.98, Support: 25)

Class 1 (F1: 0.89, Support: 5)

Note: 96.67% is holdout accuracy on a small 30-record test set, which does not guarantee real-world agricultural performance.

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Understanding the Field Through Data



Soil moisture and **soil temperature** are the two most influential features in this trained model.

Feature importance indicates the model's mathematical reliance on data, not scientific causation.

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From Prototype to Real-World Intelligence

Current State (Prototype Limitations)

1. 150-record dataset limit
2. Unverified label & crop code meanings
3. No real-world farm validation
4. Predictions are not guaranteed recommendations.



Current limitation: The dataset is small, and the meanings of its class labels and crop codes need verification. This prototype is not a substitute for expert agricultural advice.

The Roadmap (Production Vision)

1. Expand dataset across diverse regions.
2. Collaborate with agricultural experts for seasonal validation.
3. Integrate real-time IoT sensors & Weather APIs.
4. Evolve into a comprehensive decision-support platform.



Our goal is not just to predict. It is to help agriculture make better-informed decisions.

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Let's Grow a Smarter Future Together.

Turning Field Data into Smarter Agricultural Decisions

Live Demo

https://agrinextera-ai.streamlit.app/

Source Code

https://github.com/Ushapriya4596/AgriNextera-AI

Project: AgriNextera AI | Presenter: Ushapriya Akyana

Thank You! Let's Build the Future of Intelligent Agriculture.

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